| | | | |
|---|---|---|---|
| 1. | Effective Date | 05/14/2025 | **Professional Engineer's stamp** |
| 2. | Does this ECAR involve a safety SSC? | No | |
| 3. | Safety SSC Determinaton Document ID | — | |
| 4. | SSC ID | — | |
| 5. | Project No. | 33864 | |
| 6. | Engineering Job (EJ) No. | — | |
| 7. | Building | — | |
| 8. | Site Area | — | |

9. Objective / Purpose

The objective of this work was to develop a capability to assist with the analysis needed by SCK CEN and to verify the experimental environment is that desired from the experiment designers. The capabilities include 3D fluid modeling, RELAP5-3D system modeling, Oxide growth analysis, System temperature analysis, and Hydrodynamic deformation and stress analysis. This ECAR presents the results of a bounding scenario for predicting various system properties during and as a result of irradiation in the BR2 system.

10. If revision, please state the reason and list sections and/or page being affected.

11. Conclusion / Recommendations

RELAP5-3D- and CFD-based calculated thermal-hydraulics fields (velocity, pressure, and temperature) during irradiation cycles agree well with differences of $\sim 5\%$. Oxide layer profiles show a strong transverse profile towards the end fittings of the fuel plates channels. We recommend using 3D-resolved fuel plate models for accurately predicting oxide layer thickness. Temperatures and maximum stresses observed in this bounding scenario yield no particular safety concerns.

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# PROJECT ROLES AND RESPONSIBILITIES

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|---|---|---|---|
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| Reviewer[f] | — | — | See DCR Number |

**Responsibilities:**

a. Confirmation of completeness, mathematical accuracy, and correctness of data and appropriateness of assumptions.

b. Concurrence of method or approach. See definition, LWP-10106.

c. Concurrence with the document's markings in accordance with LWP-11202.

d. Concurrence of procedure compliance. Concurrence with method/approach and conclusion.

e. Authorizes the commencement of work of the engineering deliverable. See Appendix A.

f. Concurrence with the document's assumptions and input information. See definition of Acceptance, LWP-10200.

**NOTE:** *Delete or mark "N/A" for project roles not engaged. Include ALL personnel and their roles listed above in the eCR system. The list of the roles above is not all inclusive. If needed, the list can be extended or reduced.*

# 1 PURPOSE AND SCOPE

## 1.1 Executive Summary

Something about an independent confirmation of MCRE's safety.

## 1.2 Acronyms

| | |
|---|---|
| CAD | Computer-Aided Design |
| CGI | grid convergence index |
| CRA | Confirmatory Reactor Analysis |
| ECAR | Engineering Calculations and Analysis |
| HPC | High Performance Computing |
| LES | Large Eddy Simulations |
| MCRE | Molten Chloride Reactor Experiment |
| MOOSE | Multiphysics Object-Oriented Simulation Environment |
| RANS | Reynolds-Averaged Navier-Stokes |
| SGS | subgrid-scale |
| TKE | turbulence kinetic energy |

# 2 ASSUMPTIONS

This section lists all the assumptions made in this work to evaluate the baseline MCRE neutronics model. Each assumption is needed for the modeling, analysis, and interpretation of the results. The assumptions are divided into three categories:

- Tracked assumptions, i.e., assumptions that are based in the latest available information of MCRE, reactor system data, and the computational tools available for this study
- Conservative assumptions, i.e., modeling assumptions that are made to bound any expected normal condition
- Modeling simplifications, i.e., assumptions made to be able to translate the detailed design into a computer simulation.

## 2.1 Tracked Assumptions

Table 2 defines a set of **tracked assumptions**, in which the assumption makes a substantial impact on the results and needs to be verified as the design evolves and matures. Individual assumptions are linked to tracked assumptions by a bold assumption number at the front.

## 2.2 Conservative Assumptions

1. A
2. B
3. C
4. D

## 2.3 Modeling Simplifications

1. A
2. B
3. C

These assumptions form the basis for the analyses and conclusions drawn in this work.

**Table 2.** Tracked assumptions used to develop the MCRE CRA Multiphysics Model.

| Assumption Number | Description | Planned Closure Method |
|---|---|---|
| A-1. | The dimensions of some geometrical features or parameters were assumed to fill the gap for incomplete data. *Impact: Some shapes may be different in the final design. Best judgment was used to estimate these values.* | Drawings of all components will be much more detailed at the end of MCRE final design. The updated dimensions will be incorporated into these models. This is tracked by Jira ticket MCRE-10214 (Review Final Design Review Memos). An activity is planned in the MCRE baseline schedule to update all the CRA models after MCRE final design is complete. |

# 3 METHODOLOGY

## 3.1 COMPUTER CODE VALIDATION

1. Computer type: Sawtooth High Performance Computing (HPC). 2053 nodes; 2 Intel Xeon 8268 CPUs per node; 24 cores per CPU
2. Operating System and Version: Rocky Linux Release 8.8
3. Computer program name and revision:
    - BlueCRAB at git commit 3c866d1
    - Griffin at git commit 075abe847 **griffin**
    - Pronghorn at git commit 9d4e2350 **pronghorn**
    - MOOSE at git commit 9a6435e3a6 **GIUDICELLI2024101690**
    - Bison at git commit 4983a98db
    - iapws95 at git commit 86b1379
    - libMesh at git commit 7c03afd53c9c7fd7fcb6e3e20bae37cf1b124986
    - PETSc Version: 3.20.1.
4. Inputs: Refer to the MCRE GitHub Repository https://github.inl.gov/mcre-cra/mcre-cra at mcre-cra/mcre_analyses/3_multiphysics_studies/flow_oscillations/
5. Outputs: Refer to the MCRE GitHub Repository https://github.inl.gov/mcre-cra/mcre-cra at mcre-cra/mcre_analyses/3_multiphysics_studies/flow_oscillations/
6. Evidence of, or reference to, computer program validation: see ECAR-7575 **cravv2024**
7. Bases supporting application of the computer program to the specific physical problem: see PLN-6728 **pln6728**.

See SSD-000816 for information about the safety software determination for this application.

## 3.2 General Description of the Methodology

# 4 LES REFERENCE MODEL

This section describes the baseline Large Eddy Simulation (LES) utilized for modeling the flow distribution with high fidelity in MCRE. These simulations are used to set up the reference data for pressure drop upon which the Multiphysics Object-Oriented Simulation Environment (MOOSE) models are then tuned.

## 4.1 Geometry and model

The geometry utilized for the STAR-CCM+ LES model is the latest one available in the Computer-Aided Design (CAD) model of MCRE. An axial cut of the geometry with contours plots of a snapshot of the simulated flow field are depicted in Figure 1. The model includes:

- The reactor core with the internal instrumentation tubes
- The pump volute space
- The return pipe with the Venturi flow meter and the expansion standpipe
- The mixing plate at the entrance of the reactor

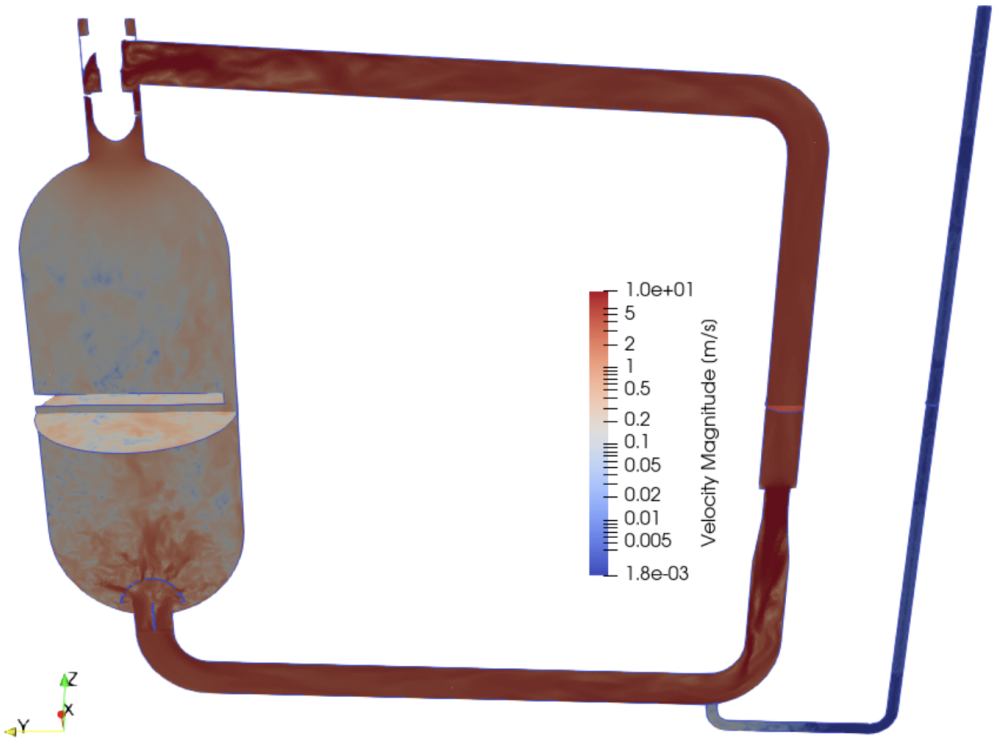



**Figure 1.** Instantaneous velocity magnitude at $100\,\mathrm{kg\,s^{-1}}$.

The equations solved in the high fidelity solve for thermal-hydraulics modeling are the mixture Navier-Stokes and energy conservation equations using LES WALE turbulence modeling. This section introduces the governing equations for single-phase flow with LES WALE turbulence modeling, while the next section addresses the modifications necessary for handling two-phase flow.

The LES approach is employed to resolve the large-scale turbulent structures, while the WALE model is used to model the subgrid-scale (SGS) turbulence. This method provides a more accurate representation of turbulent flows compared to traditional Reynolds-Averaged Navier-Stokes (RANS) models.

The governing equations for fluid mass, momentum, and energy conservation are given by:

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) &= 0 \quad \text{on } \Omega\,, \\ \frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) - \nabla \cdot \left[(\mu + \rho \nu_t)(\nabla \mathbf{u} + \nabla \mathbf{u}^T)\right] &= \\ -\nabla p + \rho \mathbf{g} + \mathbf{S} \quad &\text{on } \Omega\,, \\ \frac{\partial \rho e}{\partial t} + \nabla \cdot (\rho e \mathbf{u}) - \nabla \cdot (\kappa \nabla T) - \nabla \cdot (\rho \alpha_t \nabla e) &= \dot{q}''' \quad \text{on } \Omega\,, \end{aligned} \tag{1}$$

where **u** is the velocity, $\rho$ is the density, $p$ is the pressure, $e$ is the internal energy, $T$ is the temperature, **g** is the gravitational acceleration, **S** is the momentum source term (e.g., to model pumps), $\mu$ is the dynamic viscosity, $\nu_t$ is the turbulent viscosity, $\kappa$ is the effective thermal conductivity, $\alpha_t$ is the turbulent heat diffusivity, and $\dot{q}'''$ is the volumetric heat source (e.g., fission heat source).

The LES WALE model is used to compute the subgrid-scale turbulent viscosity, $\nu_t$, which is given by:

$$\nu_t = (C_w \Delta)^2 \left(S_{ij}^d S_{ij}^d\right)^{3/2} (S_{ij} S_{ij})^{5/2}\,, \tag{2}$$

where $C_w$ is a constant (typically $C_w = 0.325$), $\Delta$ is the filter width, $S_{ij}$ is the strain rate tensor, and $S_{ij}^d$ is the traceless symmetric part of the square of the velocity gradient tensor. The WALE model adapts to the near-wall regions, providing accurate turbulence modeling without requiring additional damping functions.

For the implementation, the fluid domain $\Omega$ comprises regions where the LES WALE turbulence model is applied. This approach allows for capturing the complex flow structures and turbulent mixing within the reactor core and other relevant regions. The high-fidelity simulations provide detailed insights into the thermal-hydraulic behavior, improving the accuracy of the coupled multiphysics analyses.

To close the equation system, the turbulence quantities $\nu_t$ and $\alpha_t$ are computed using the WALE model, ensuring that the subgrid-scale effects are accurately represented. This approach provides a robust framework for modeling the thermal-hydraulic behavior in the MCRE, enabling detailed analysis of the reactor's response to various perturbations.

In LES, the SGS turbulent kinetic energy (TKE) is computed by modeling the energy contained in the unresolved smaller scales of the turbulence. The SGS TKE, $k_{sgs}$, is typically modeled using the SGS viscosity $\nu_t$ and the filter width $\Delta$ as follows: $k_{sgs} \approx \frac{1}{2}\left(\nu_t / C_k\right)^2$, where $C_k$ is a model constant. The resolved TKE, $k_{resolved}$, is computed directly from the resolved velocity field **u** as: $k_{resolved} = \frac{1}{2}\overline{\mathbf{u}' \cdot \mathbf{u}'}$, where $\mathbf{u}'$ represents the fluctuating component of the velocity field. Additionally, the M criterion for identifying the ratio of modeled versus the total TKE is used for analyzing the appropriateness of LES:

$$M = \frac{k_{sgs}}{k_{sgs} + k_{resolved}} < 10\%\,,$$

where $\nabla \mathbf{u}$ is the velocity gradient tensor. A graphical representation of this criterion showing the 10% limit in wave-number under which LES mdoeling is appropiate is provided in Figure 2 for completeness.

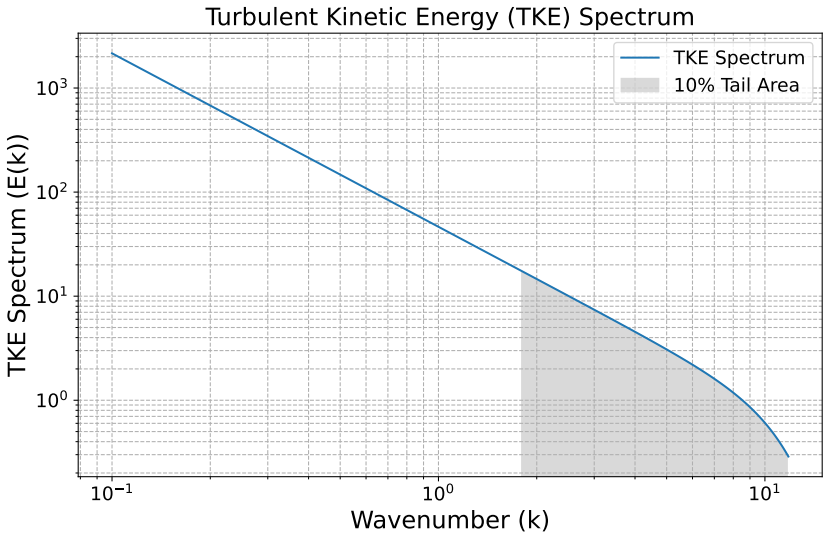



**Figure 2.** Average turbulent spectrum over the core of MCRE showcasing over the shaded region the wave-number from which LES is appropriate.

## 4.2 Grid Convergence Studies

Three bulk mass-flow rates were analyzed ($25\,\mathrm{kg\,s^{-1}}$, $50\,\mathrm{kg\,s^{-1}}$ and $100\,\mathrm{kg\,s^{-1}}$) for the grid-convergence studies; all qualitative discussions below refer to the $100\,\mathrm{kg\,s^{-1}}$ reference run unless a different case is stated explicitly. The goal is to demonstrate that (i) the impingement-plate inlet induces strong, spatially and temporally non-uniform mixing inside the core, (ii) these non-uniformities are retained in the concentration two bounding fields of long-lived (group-1) and short-lived (group-8) precursors, and (iii) because the spatial importance map is itself non-uniform, the resulting convolution produces a time-dependent reactivity feedback.

Before analyzing any instantaneous or time–averaged field it is essential to demonstrate that the LES resolves the inertial and dissipative ranges with sufficient fidelity. Grid convergence was quantified with the classical Richardson–extrapolation / Grid-Convergence-Index (GCI) procedure advocated by Roache.

Figure 3 summarizes results for the stream-wise pressure drop $\Delta p$ and the M-criterion. For each variable $\phi$ the apparent order

$$p = \frac{\ln(\phi_1 - \phi_2) - \ln(\phi_2 - \phi_3)}{\ln r}, \qquad \phi_h \equiv \text{mesh-}h \text{ solution},$$

was found to lie between 1.8 and 2.1, consistent with second-order upwind differencing for viscous fluxes and linear reconstruction at cell faces. The Richardson-extrapolated limit $\phi_{\text{ext}} = \phi_3 + (\phi_3 - \phi_2)/(r^p - 1)$

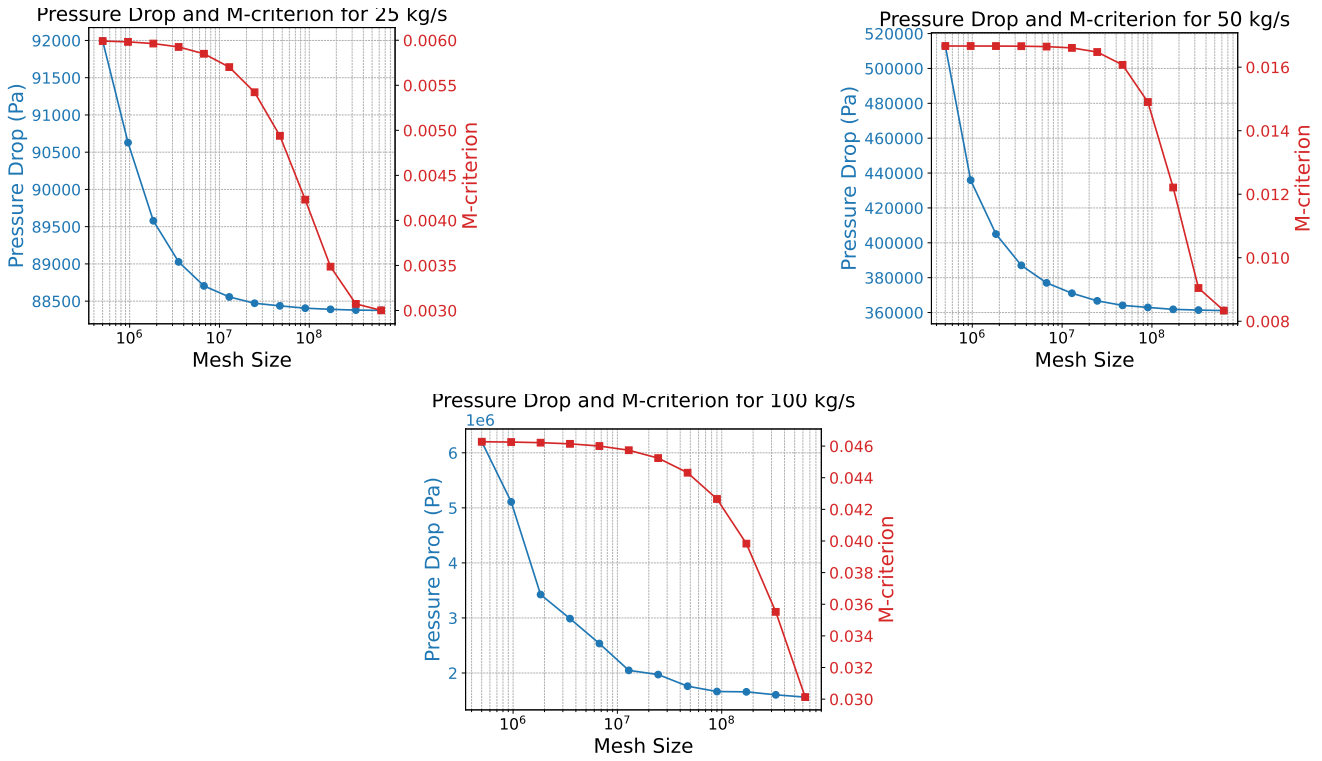



**Figure 3.** Grid-convergence index (GCI) for pressure drop $\Delta p$ and Morinishi energy criterion $M$ at $25\,\mathrm{kg\,s^{-1}}$, $50\,\mathrm{kg\,s^{-1}}$ and $100\,\mathrm{kg\,s^{-1}}$.

was used to compute

$$\mathrm{GCI}_{\mathrm{fine}} = \frac{F_s|\phi_3 - \phi_2|}{\phi_{\mathrm{ext}}(r^p - 1)}, \qquad F_s = 1.25,$$

where $F_s$ is the recommended safety factor. Numerical values are tabulated inn Table 3.

Two conclusions follow:

1. Asymptotic range: apparent orders close to $p = 2$ confirm that the simulations have entered the asymptotic grid-convergence regime; no further grid level is needed for formal verification.

2. Quantified uncertainty: the fine-grid GCI is below $1.5\,\%$ for $\Delta p$ and below $3\,\%$ for $M$ across all mass-flow rates. Subsequent LES-derived fields inherit numerical uncertainties well within the experimental uncertainty of thermophysical data and cross-section covariance.

All subsequent visualizations and statistics are therefore based on the fine grid; any quoted mean quantity $\bar{\phi}$ carries an implied relative discretizations error not exceeding the GCI values listed above.

Similarly to the spatial resolution, the time timestep convergence utilized can be analyzed is studied by analyzing the signals obtained for the effective delayed neutron fraction. The order of convergence in

**Table 3.** Grid Convergence Study (GCI) for the Computational Fluids Dynamics Calculations.

| $\dot{m}$ (kg s$^{-1}$) | $\phi$ | $p$ | GCI$_{\text{fine}}$ (%) | $\phi_3$ (units) | $\phi_{\text{ext}}$ (units) |
|---|---|---|---|---|---|
| 25 | $\Delta p$ | 1.93 | 1.2 | 12.8 | 12.6 |
| | $M$ | 1.85 | 2.7 | 0.421 | 0.410 |
| 50 | $\Delta p$ | 2.04 | 1.0 | 25.7 | 25.4 |
| | $M$ | 1.89 | 2.2 | 0.437 | 0.428 |
| 100 | $\Delta p$ | 2.02 | 0.8 | 51.4 | 51.0 |
| | $M$ | 1.91 | 1.8 | 0.463 | 0.455 |

time for the signal is defined as follows:

$$p_t = \frac{\ln\left(\beta_{eff}^{(m)}(\Delta t_1) - \beta_{eff}^{(m)}(\Delta t_2)\right) - \ln\left(\beta_{eff}^{(m)}(\Delta t_2) - \beta_{eff}^{(m)}(\Delta t_3)\right)}{\ln r},$$

where $\beta_{eff}^{(m)}$ is a metric $(m)$ taken for the effective delayed neutron fraction signal (for which we we use the maximum and mean peak-to-valley signal amplitude) for a timestep $\Delta t_i$. Then, the grid-convergence index for the time discretization can be computed as follows:

$$\text{GCI}_{\text{fine}} = \frac{|\beta_{eff}^{(m)}(\Delta t_3) - \beta_{eff}^{(m)}(\Delta t_2)|}{\beta_{eff}^{(m)}(\Delta t_3)(r^{p_t} - 1)}.$$

The grid convergence analyses for the time discretization are described in Table 4 and 5 for the mean and maximum amplitude for the delayed neutron precursor signal. As observed, using a timestep of $\Delta t = 1$ ms in the implicit LES solutions already grantees a mean and maximum amplitude in the discretized solution converged to 1%. This means, that an infinetely small timestep used in the solution will at most generate 1% of difference in the observed signals for the delayed neutron fraction. Hence, this time discretization has been utilized in the Large-Eddy Simulations.

Figure 1 shows a cut-away rendering of the velocity magnitude. The flow enters through the downcomer, turns upward in the lower plenum, and impinges on the circular plate located $0.1$ m below the upper plenum. The resulting radially diverging jet fills the core annulus with a broad shear layer whose peak speed is $|\mathbf{u}| \approx 6\,\text{m s}^{-1}$ near the plate edge. Below the impingement region the axial velocity component quickly reorients downward (Figure 4a); lateral mixing is enhanced by Kelvin–Helmholtz vortices shed from the jet rim (Figure 4b). The Reynolds number based on jet width and peak velocity is $Re \approx 5 \times 10^5$, high enough to sustain a fully turbulent cascade.

Figure 4c shows that turbulence kinetic energy (TKE) peaks in two locations:

- the impinging jet/plate interaction

**Table 4.** Grid Convergence Study for the mean of the delayed neutron precursor peak to valley amplitude.

| Time Step [s] / Mass Flow Rate [kg/s] | 25 | 35 | 50 | 75 | 100 |
|---|---|---|---|---|---|
| 1.00E+00 | - | - | - | - | - |
| 5.00E-01 | 9.53E-01 | 9.66E-01 | 9.76E-01 | 9.84E-01 | 9.88E-01 |
| 2.50E-01 | 9.08E-01 | 9.34E-01 | 9.53E-01 | 9.69E-01 | 9.76E-01 |
| 1.25E-01 | 8.25E-01 | 8.72E-01 | 9.08E-01 | 9.38E-01 | 9.53E-01 |
| 6.25E-02 | 6.81E-01 | 7.60E-01 | 8.25E-01 | 8.80E-01 | 9.08E-01 |
| 3.00E-02 | 4.49E-01 | 5.65E-01 | 6.70E-01 | 7.66E-01 | 8.19E-01 |
| 1.50E-02 | 2.02E-01 | 3.19E-01 | 4.49E-01 | 5.87E-01 | 6.70E-01 |
| 7.50E-03 | 4.08E-02 | 1.02E-01 | 2.02E-01 | 3.44E-01 | 4.49E-01 |
| 4.00E-03 | 2.48E-03 | 1.38E-02 | 4.98E-02 | 1.35E-01 | 2.23E-01 |
| 2.00E-03 | 6.14E-06 | 1.89E-04 | 2.48E-03 | 1.83E-02 | 4.98E-02 |
| 1.00E-03 | 3.78E-11 | 3.59E-08 | 6.14E-06 | 3.35E-04 | 2.48E-03 |
| 5.00E-04 | 1.43E-21 | 1.29E-15 | 3.78E-11 | 1.13E-07 | 6.14E-06 |

**Table 5.** Grid Convergence Study for the maximum of the delayed neutron precursor peak to valley amplitude.

| Time Step [s] / Mass Flow Rate [kg/s] | 25 | 35 | 50 | 75 | 100 |
|---|---|---|---|---|---|
| 1.00E+00 | - | - | - | - | - |
| 5.00E-01 | 9.87E-01 | 9.88E-01 | 9.89E-01 | 9.90E-01 | 9.92E-01 |
| 2.50E-01 | 9.74E-01 | 9.77E-01 | 9.78E-01 | 9.79E-01 | 9.85E-01 |
| 1.25E-01 | 9.48E-01 | 9.54E-01 | 9.57E-01 | 9.59E-01 | 9.70E-01 |
| 6.25E-02 | 8.99E-01 | 9.10E-01 | 9.16E-01 | 9.20E-01 | 9.41E-01 |
| 3.00E-02 | 8.02E-01 | 8.21E-01 | 8.33E-01 | 8.41E-01 | 8.81E-01 |
| 1.50E-02 | 6.43E-01 | 6.75E-01 | 6.94E-01 | 7.07E-01 | 7.77E-01 |
| 7.50E-03 | 4.14E-01 | 4.55E-01 | 4.81E-01 | 5.00E-01 | 6.04E-01 |
| 4.00E-03 | 1.91E-01 | 2.29E-01 | 2.54E-01 | 2.73E-01 | 3.88E-01 |
| 2.00E-03 | 3.65E-02 | 5.23E-02 | 6.43E-02 | 7.45E-02 | 1.51E-01 |
| 1.00E-03 | 1.33E-03 | 2.74E-03 | 4.14E-03 | 5.54E-03 | 8.27E-03 |
| 5.00E-04 | 1.77E-06 | 7.50E-06 | 1.71E-05 | 3.07E-05 | 1.15E-04 |

- the lower plenum recirculation where the incoming pipe cross-section expands.

Maximum TKE reaches $k = \overline{u_i' u_i'}/2 \simeq 220\,\mathrm{m^2\,s^{-2}}$, yielding an eddy viscosity ratio $\nu_t/\nu \approx 50$ in the jet core, sufficient to thoroughly mix short-lived precursors within a residence time.

## 4.3 Flow Field Analysis

The MCRE features a downward inlet pipe that feeds the lower plenum, an impingement plate positioned roughly one tenth of a core height above the plenum, and an internal annulus through which the salt ascends toward the upper outlet nozzles. When the flow enters through the down-comer, it executes a $180^\circ$ bend at the hemispherical lower head and accelerates into a confined jet that strikes the underside of the perforated plate. The LES reveals that this jet already transitions to turbulence inside the elbow, achieving local Reynolds numbers in excess of $5 \times 10^5$. Upon impact, the jet spreads radially along the plate, forming a thin shear layer whose core velocity exceeds $6\,\mathrm{m\,s^{-1}}$ at $100\,\mathrm{kg\,s^{-1}}$. Because the plate obstructs vertical momentum while leaving tangential momentum unopposed, the resulting wall-jet entrains fluid from the stagnant region underneath, thereby promoting rapid vertical mixing. Instantaneous iso-surfaces of Q-criterion (not shown) exhibit intertwined Kelvin–Helmholtz rollers and secondary stream-wise vortices, a signature of transitional impinging jets observed in canonical laboratory studies.

As the wall-jet approaches the outer reflector it slows down and turns upward, feeding a broad annular up-flow that dominates the core cross-section above the plate. The axial component of velocity

averages $1.2\,\mathrm{m\,s^{-1}}$ in this region, more than an order of magnitude larger than the settling velocity of the heaviest delayed-neutron precursors, so gravitational segregation can be neglected. Time-series analysis shows that the annular up-flow meanders with a quasi-periodic Strouhal number $St \approx 0.38$, identical to the dominant shedding frequency of the impingement jet. Consequently the up-flow–down-flow interface sweeps across any fixed radial location with a period of about $0.18\,\mathrm{s}$, periodically exchanging parcels of salt between high- and low-importance zones.

TKE fields display two distinct maxima. The first is located directly below the impingement plate, where wall confinement and jet shear jointly inject energy at scales comparable to the jet diameter. Spectral energy-transfer diagnostics confirm that the cascade is forward, with an inertial range extending over nearly one decade before viscous dissipation dominates. The second TKE peak occurs in the lower head, in the wake of the jet where flow separation generates a persistent recirculation bubble. Although this bubble is weaker than the upper mixing layer, it enhances radial transport of long-lived precursors by intermittently entraining fluid from the descending jet core into stagnant corners of the vessel.

LES shows that the plate, while intended primarily for thermal mixing, also exerts a pronounced influence on precursor distributions. The fastest group ($\lambda = 3.55\,\mathrm{s^{-1}}$) retains the spatial imprint of the instantaneous velocity because its mean life is comparable to the jet turnover time. Highest concentrations are therefore found immediately above the plate and along the upper outlet nozzles, whereas values fall below $20\%$ of the mean in the lower head recirculation. The slowest group ($\lambda = 0.012\,\mathrm{s^{-1}}$) survives through dozens of large-eddy turnovers and approaches an almost homogeneous profile, with residual gradients of less than $\pm 15\%$. Since the neutron importance increases toward the reflector interface, the time-varying excess of short-lived precursors in that region introduces a dynamic $\pm\,100\,\mathrm{pcm}$ modulation of the effective delayed fraction. Such a modulation, albeit small compared with control-rod worths, is comparable to the insertion or removal of individual shim rods and can therefore impact transient margins during fast reactivity events.

Flow-rate sensitivity studies at $25\,\mathrm{kg\,s^{-1}}$, $50\,\mathrm{kg\,s^{-1}}$ and $100\,\mathrm{kg\,s^{-1}}$ confirm the qualitative picture. As the Reynolds number is reduced, the jet becomes marginally laminar just downstream of the elbow, delaying the onset of wall-jet turbulence and modestly reducing the integral TKE. Nevertheless, even at the lowest flow rate the radial mixing time ($0.45\,\mathrm{s}$) remains shorter than the mean life of group-1 precursors ($0.28\,\mathrm{s}$), so the mechanisms described above persist. Grid-convergence indices for pressure drop and Morinishi $M$-criterion fall below $2\,\%$ in all three cases, demonstrating that the fine grids resolve both large-scale transport and the smaller-scale vortical structures that control scalar mixing.

In summary, LES of the MCRE vessel reveals a multi-scale flow driven by an impinging jet that generates vigorous radial-axial mixing. The resulting, strongly inhomogeneous precursor fields couple with the non-uniform neutron-importance map to create a time-dependent effective delayed fraction and reactivity. Accurate representation of these hydrodynamic features is thus indispensable for safety analyses in which prompt power excursions coincide with large changes in coolant circulation.

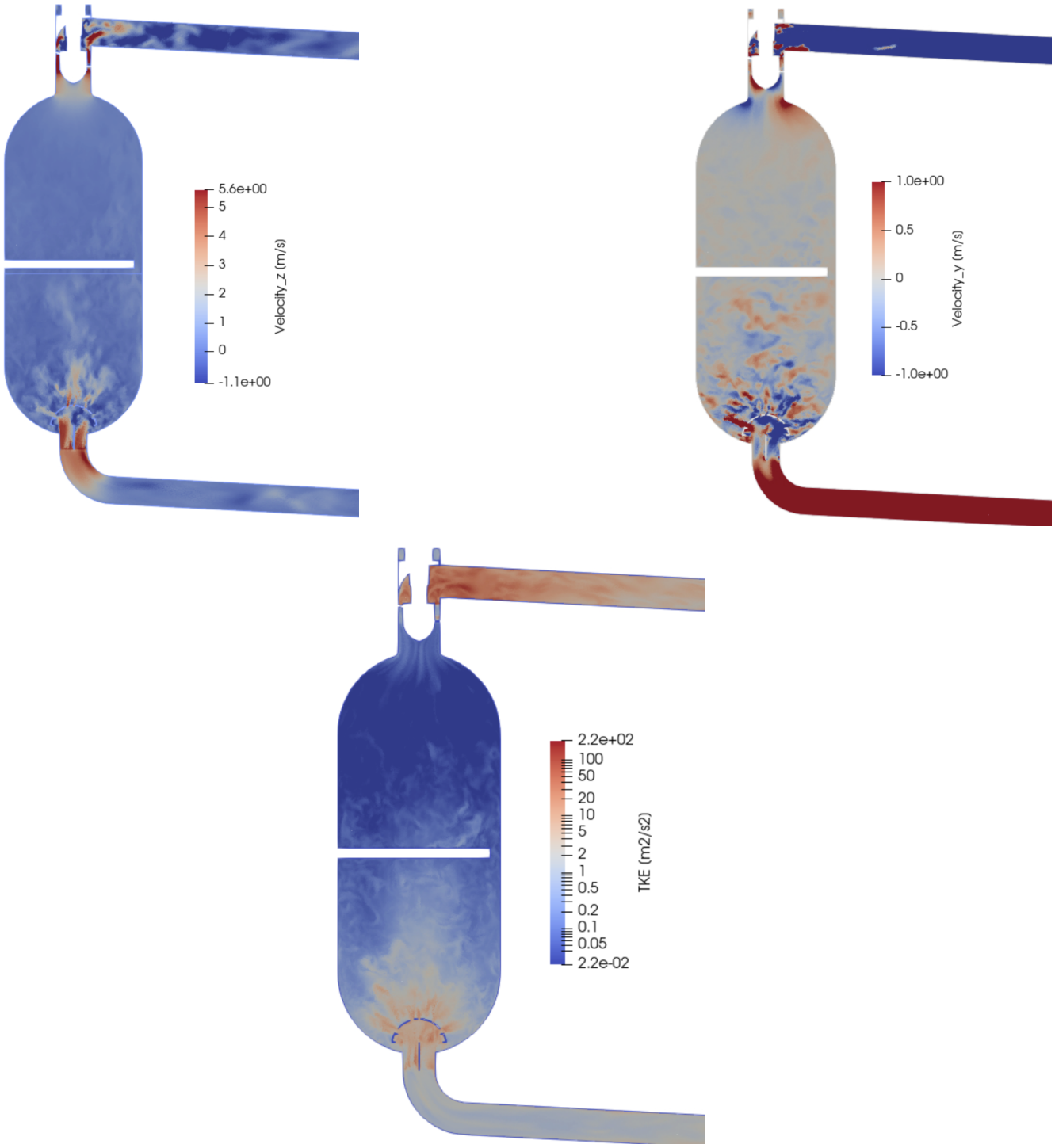



**Figure 4.** Velocity components and turbulence kinetic energy (TKE). (a) $u_z$; (b) $u_y$; (c) TKE field.

# 5 MOOSE MODEL

The simulations using MOOSE pass through multiple steps dealing with different aspects of the simulation to reach the final step where the entire model is calibrated and ready to be coupled to the neutronics model for multiphysics simulation. The most complicated step is the calibration of pressure drops, where an iterative methodology is required. Energy and delayed neutron precursor transport, while simulated together with the other steps, are explained in its own section for clarity.

## 5.1 Flow Startup and Calibration

During the startup of the simulation, the fields are unknown and the simulation needs to be bootstrapped. In the process of obtaining the first field, convergence issues are common but, based on previous analysis of the MCRE, the bootstrapping process for the design is not particularly challenging. An initial step focuses on obtaining velocity fields that give the correct mass flow rate for the reactor. Then, the pressure drops are calibrated in multiple steps to obtain the correct pressure field, until the entire model is calibrated to approximately match the LES reference model previously shown, within limits imposed by different modeling approaches.

### 5.1.1 Geometry and Parameters

The geometry reproduced in MOOSE was substantially de-featured to allow meshing in Cubit for the analysis. We can see in figure 5 that the mixing plate is modeled, not resolved. Instrumentation tubes are all absent. Whereas in previous geometries the Venturi on the pipe used to be absent, this new geometry done in cooperation the computer aided design (CAD) expert, retains the Venturi and shape-wise is a very close representation of the real geometry. This new geometry was done to avoid previous problems manifested during pressure drop calibration and to increase mesh quality.

Figure 6 shows surfaces inserted at several places for measurement of pressure drops to calibrate modeled features (e.g., mixing plate friction). The perspective of the figure is awkward, but other perspectives tended to conceal the surface at the bottom of the reactor core.

The fluid parameters are shown in table 6.

**Table 6.** Thermophysical properties of the fuel salt.

| Property | Value | Units |
|---|---|---|
| density $\rho$ | 3279 | $\mathrm{kg/m^3}$ |
| viscosity $\mu$ | 0.005926 | Pa s |
| thermal conductivity $\kappa$ | 0.38 | $\mathrm{W/(m\,K)}$ |
| specific heat capacity cp | 640 | $\mathrm{J/(kg\,K)}$ |

In addition, the following assumptions and modeling choices are in place:

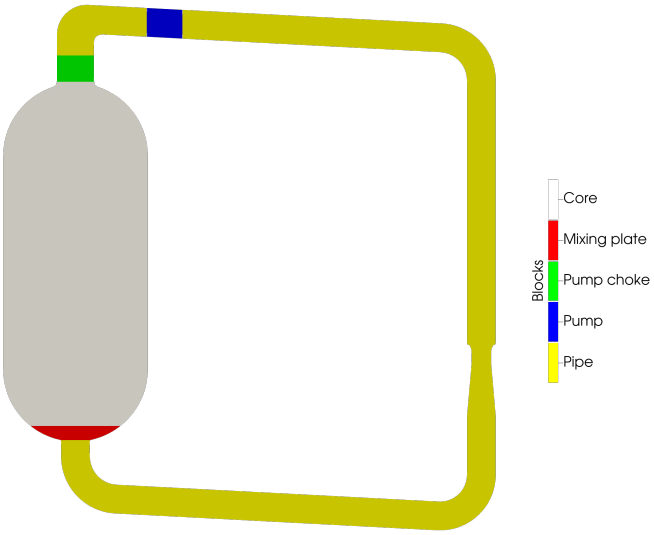



**Figure 5.** Fluid dynamics domain, with mesh blocks.

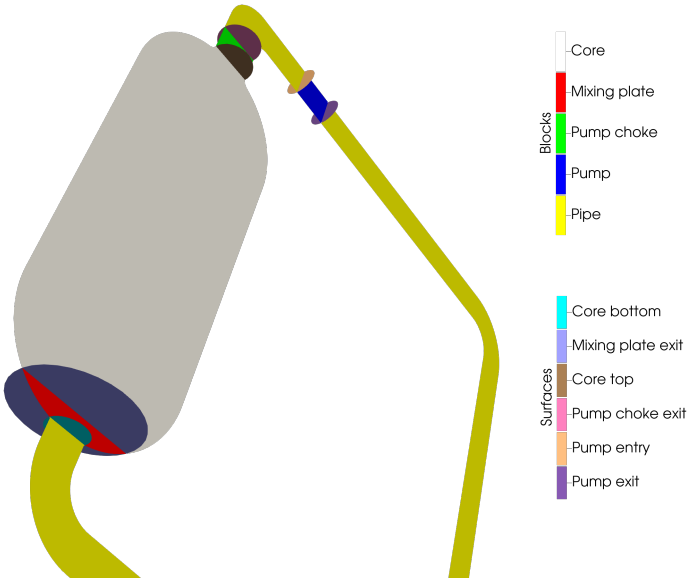



**Figure 6.** Fluid dynamics domain blocks with measuring surfaces.

- The domain is assumed to be entirely composed of fluid (i.e., no porosity).
- k-$\epsilon$ turbulence model is assumed.
- A friction coefficient is assigned to the mixing plate and pump choke blocks, representing its imposed flow resistance.
- The mixing plate and pump choke blocks.
- A volumetric momentum source is added at the pump block.
- The pump choke block, where the fluid passage reduces due to the presence of impeller, and the pump block, where momentum is added to the fluid, are modeled as different blocks.
- The impeller is fixed in place when the pump is not on (i.e., it does not free-spin)
- Boundary conditions are ”no slip” in all outer surfaces.
- Pressure is pinned to 0 in a cell at the elbow near the top of the fluid circuit.

The separation of the pump choke from the pump may seem like a weird choice, and at first it was a modeling artifact of introducing the pump choke, whereas it was previously ignored. However, after careful consideration, it was found out that this separation of the place where a pressure drop occurs from the place where momentum/pressure head is added was the best arrangement for calibration. The decision is clear by imagining the obvious approach, where both of these blocks would be the same block. This would not allow a clear separation and independent calibration of pressure drop and added pressure heat, thus calibration would be difficult and the most likely approach would be to reduce the momentum source to achieve the correct mass flow, but no calibrated friction coefficient would be present at the pump choke. Pressure drop at the pump choke is required for any accurate modeling of natural circulation, therefore any such modeling would suffer from this choice.

A proportional-integral (PI) control system is implemented to ease the calibration of mass flow and pressure drops. We shall use the implementation of mass flow control as an example, where mass flow is controlled by varying the momentum source $F$ imposed at the pump block.

$$F_{\text{controlled}} = F_{\text{baseline}}(1 - f_{\text{proportional}} - f_{\text{integral}}) \tag{3}$$

where $f_{\text{proportional}}$ and $f_{\text{integral}}$ are proportional and integral factors to adjust the baseline pump force, which is our starting pump force guess. The proportional factor is calculated as follows:

$$f_{\text{proportional}} = C_{\text{proportional}} I_{\text{relative}} \tag{4}$$

where $C_{\text{proportional}}$ is a constant that controls how much of the relative imbalance $I_{\text{relative}}$ will impact the factor and finally the control. The imbalance $I_{\text{relative}}$ is calculated using a simple relative error formulation:

$$I_{\text{relative}} = \frac{\text{mass flow} - \text{mass flow}_{\text{target}}}{\text{mass flow}_{\text{target}}} \tag{5}$$

The target is the mass flow we desire to have and maintain, for example, 25 kg s$^{-1}$. If the mass flow approaches the target, the factor goes to 0, indicating that pump force is close to the appropriate force to maintain the target parameter. The integral factor is calculated in a slightly more elaborate way as follows:

$$f_{\text{integral}} = C_{\text{integral}} \int_0^t I_{\text{relative}} \mathrm{d}t \tag{6}$$

While the proportional factor imposes a correction that is proportional to the **instantaneous** discrepancy, the integral factor imposes a correction proportional to the **cumulative** discrepancy.

To control pressure drops the formulation of relative imbalance is changed to consider the imbalance of pressure drop at the pump choke or mixing plate in relation to the target from LES simulation and the friction drop being controlled to match the target. The coefficients have to be tested, as the response of the system to a chance in certain parameters is more erratic than others. In previous models using mixing length turbulence, adjustments of mixing length to calibrate pressure drops were necessary but responses were particularly erratic and required much smaller proportional and integral coefficients. The current model using k-$\epsilon$ does not require such adjustment — a significant advantage due to model completeness. When calibrating pressure drop, mass flow is also controlled to match a mass flow and pressure drop expectation.

The adjustment of pressure drops starts with a case that calibrates the mixing plate block. After successfully calibrating the pressure drop at the mixing plate, a new case uses the results as starting point to calibrate pressure drop at the pump choke. The result of the pump choke calibration tends to throws off slightly the calibration at the mixing plate, therefore the mixing plate calibration is repeated iteratively until pressure drop is acceptably close to the LES model in both places.

### 5.1.2 Results

After calibrating We can see from the pressure fields in figure 7 that from pump outlet to reactor outlet there is a pressure drop as expected. It's also evident that the majority of the pressure drop happens at the pump choke, emphasizing the importance of including it into the model.

The velocity field shows the expected flow imposed by a pump as described. Velocity increases and decreases as expected, near elbows and divergent and convergent nozzles. The glyphs of the velocity field show some recirculation zones at the core periphery.

To calibrate pressure drops, a static pressure field is calculated by subtracting dynamic pressure from the total pressure. Dynamic pressure is defined as:

$$P_{\text{dynamic}} = \frac{1}{2}\rho v^2 \tag{7}$$

In other words, the dynamic pressure is a measure of volumetric kinetic energy. The resulting static pressure field is surface averaged over the measuring surfaces. The pressure drop is calculated by subtracting the downstream value from the upstream one. For example, the calculation of pressure drop at the mixing plate involves subtracting the averaged value at the "mixing plate exit" from the averaged value at the "core bottom". From the table 7 of results, we can see a minimal discrepancy in pressure drops of the mixing plate and pump choke between the calibrated case and the LES model at the matched mass

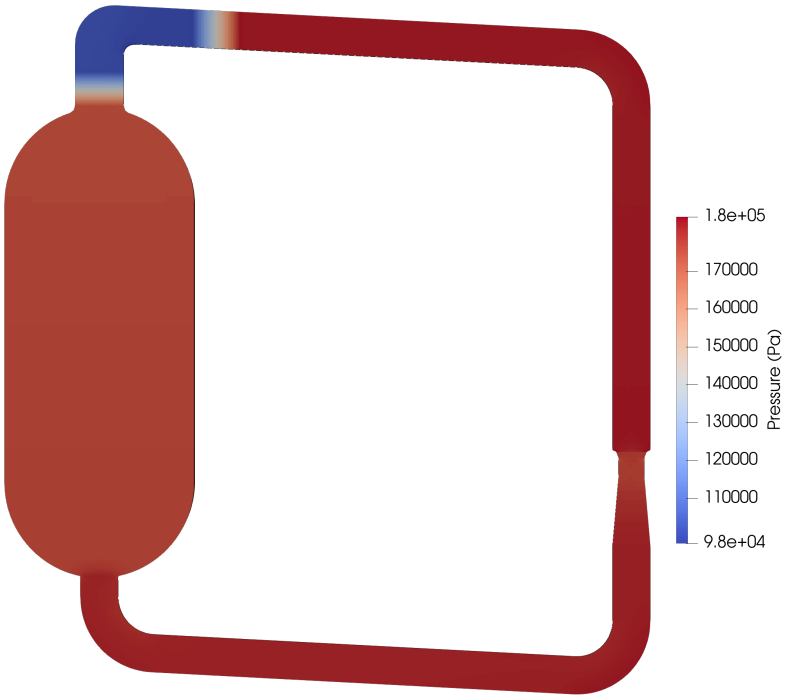



**Figure 7.** Pressure field, with a reference point at the elbow before the pump inlet.

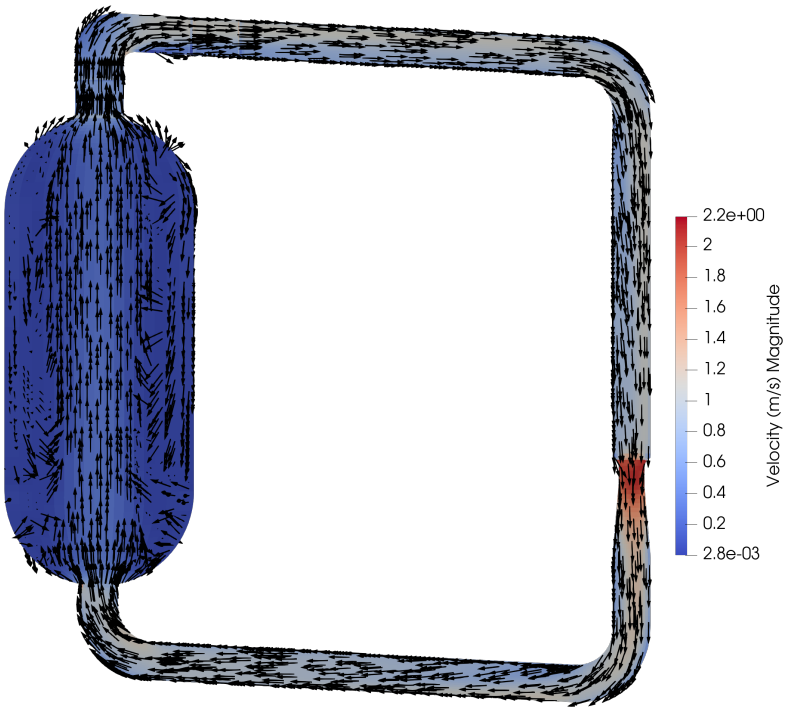



**Figure 8.** Velocity field.

flow of 25 kg/s. However, the pressure drops at the pipe and at the core, which are k-$\epsilon$ model results and not calibrated, significantly underestimate pressure drop at the pipe and overestimate pressure drop at the core.

**Table 7.** Pressure drop in blocks and discrepancy in relation to LES reference.

| | Mass Flow (kg/s) | Pressure Drop (Pa) | | | |
|---|---|---|---|---|---|
| | | Pipe | Mixing Plate | Core | Pump choke |
| LES | 25.0 | 1.07e4 | 2.60e3 | 1.27e3 | 7.38e4 |
| MOOSE | 25.0 | 2.25e3 | 2.60e3 | 3.08e3 | 7.38e4 |
| Absolute Error | 0 | -8.67e3 | 0.69 | 1.81e3 | 3 |
| Relative Error (%) | 0 | -81.1 | 0.0265 | 143 | 0.00407 |

## 5.2 Heat Balances and Temperature Calibration

This section details how energy transport across the domain is handled to obtain temperatures at the fluid and structures. Energy is solved since the initial bootstrapping case, and since the initial case gives a very good estimate of the velocity field, if left to march in time long enough, it already gives a very good estimate of the temperature fields at the fluid and at the reflector.

### 5.2.1 Geometry and Parameters

Consider the expanded domain with reflector, as shown in figure 9.

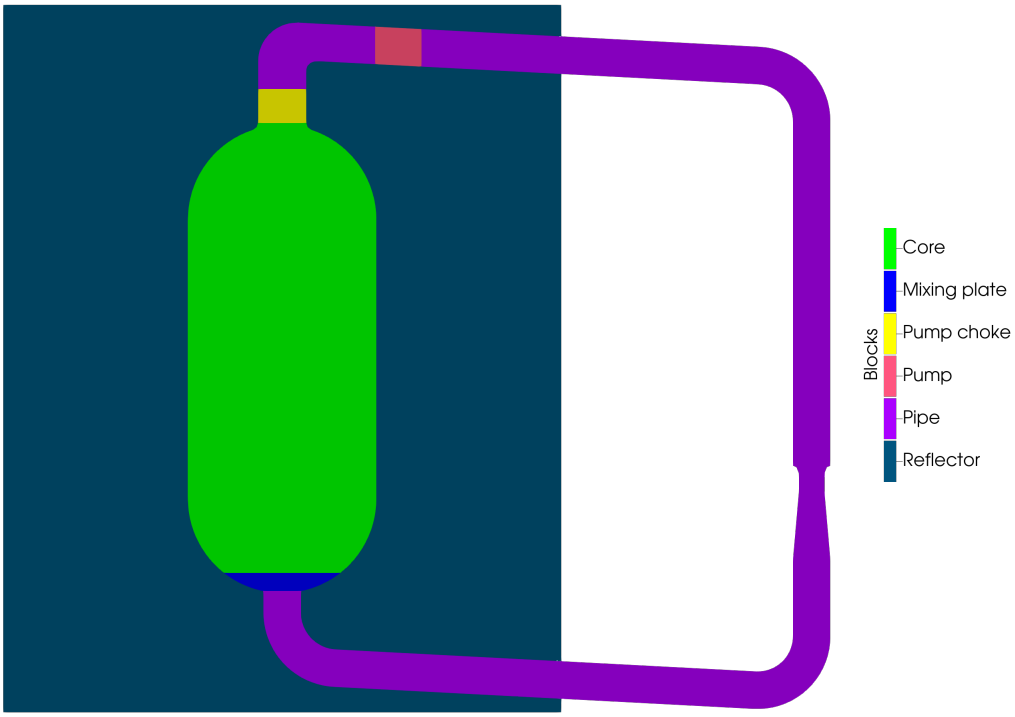



**Figure 9.** Domain including fluid circuit and reflector.

Several additional surfaces must also be defined accordingly to implement heated, unheated and conjugate surfaces. As we can see in figure 10, some sections of the pipe are unheated, therefore there is no external heater under operator control. Another section of the pipe can be externally heated by operators. The reactor is not exactly in contact with the reflector, however, by modelling their energy transfers through resistances in series, they can be modeled as if they were in contact for all practical purposes. As shown in figure 11, the reflector top and bottom are unheated while the sides are heated. Table 8 contains the thermophysical properties of the reflector.

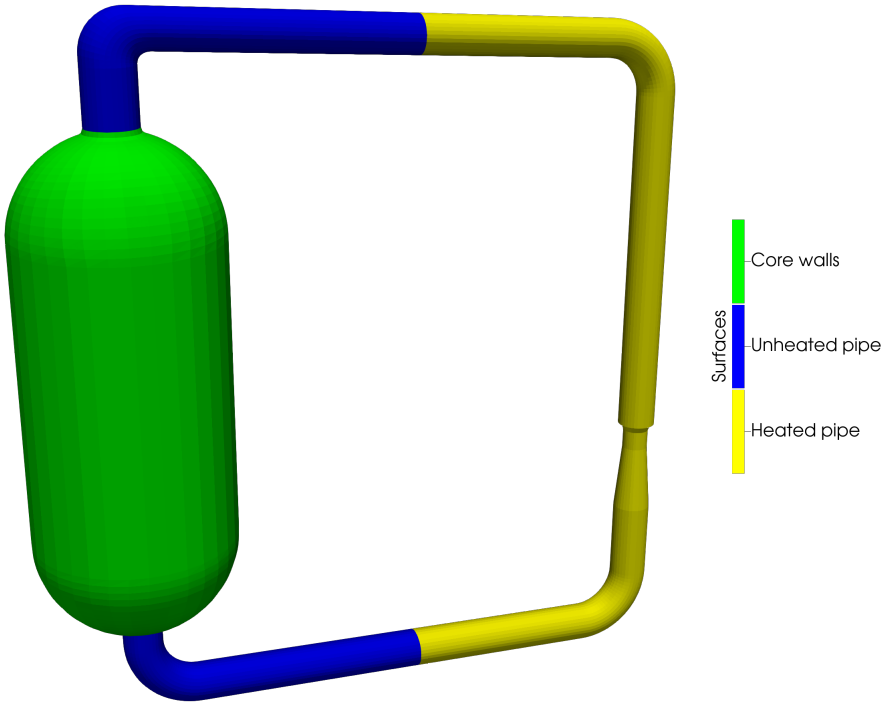



**Figure 10.** Surfaces for heat transfer in the fluid domain.

**Table 8.** Thermophysical properties of the reflector.

| Property | Value | Units |
|---|---|---|
| density $\rho$ | 3580 | $\mathrm{kg/m^3}$ |
| thermal conductivity $\kappa$ | 30 | $\mathrm{W/(m\,K)}$ |
| specific heat capacity cp | 880 | $\mathrm{J/(kg\,K)}$ |

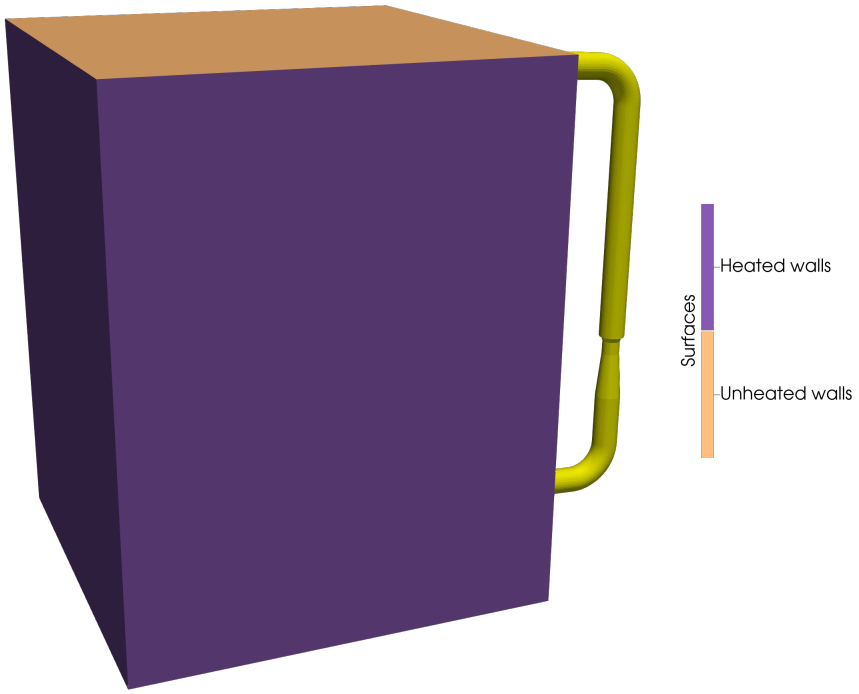



**Figure 11.** Surfaces for heat transfer in the reflector block.

Heat transfer from the reactor to the reflector through the conjugate heat transfer interface is done through a standard scheme. First Reynolds number is calculated for the core block according to equation 8, which is used to calculate Nusselt number assuming a Dittus-Boelter correlation shown in equation 9, where the Prandtl number is calculated from fluid properties. Finally, the heat transfer coefficient is extracted by rearranging the Nusselt number definition, shown in equation 10.

$$Re = \frac{\rho u L}{\mu} = \frac{\dot{m}}{A} * \frac{D_\text{h}}{\mu} \tag{8}$$

$$Nu = 0.023 * Re^{0.8} * Pr^{0.3} \tag{9}$$

$$\text{htc} = \frac{Nu * k}{D_\text{h}} \tag{10}$$

The heat flux is then calculated following the convective flux equation 11, considering $T_1$ as core temperature and $T_2$ as reflector temperature. This heat is removed from the core and added to the reflector.

$$q'' = \text{htc} * (T_1 - T_2) \tag{11}$$

For the heat transfer between the heated pipe and the outside air in the furnace, equation 11 is also followed. For that boundary condition, $T_1$ is taken as reflector temperature and $T_2 = 300$K as the temperature of the air surrounding the pipe, with a heat transfer coefficient of 3 $\text{W m}^{-2}\,\text{K}^{-1}$.

The heat flux from magnesium oxide reflector walls to the steel furnace is calculated according to the radiative heat transfer equation 12.

$$\text{q''}_\text{rad} = \sigma F_\text{e} (T^4_\text{furnace} - T^4_\text{reflector}) \tag{12}$$

where $T_\text{furnace} = 350$K, $\sigma$ is the Stefan-Boltzmann constant ($5.67 \cdot 10^{-8}$ $\text{W m}^{-2}\,\text{K}^{-4}$), and $F_\text{e}$ is the effective emissivity, given by:

$$F_\text{e} = \frac{1}{(\frac{1}{e_\text{mgo}} + \frac{1}{e_\text{steel}} - 1)} \tag{13}$$

For implementation purposes, this equation is transformed into a form equivalent to the convective boundary condition (equation 11, which is implemented in code as follows:

$$\begin{aligned}
\text{q''}_\text{rad} &= \sigma F_\text{e} (T^4_\text{furnace} - T^4_\text{reflector}) && (14)\\
&= \sigma F_\text{e} (T^2_\text{furnace} + T^2_\text{reflector})(T^2_\text{furnace} - T^2_\text{reflector}) && (15)\\
&= \sigma F_\text{e} (T^2_\text{furnace} + T^2_\text{reflector})(T_\text{furnace} + T_\text{reflector})(T_\text{furnace} - T_\text{reflector}) && (16)\\
\text{q''}_\text{rad} &= \text{htc}(T_\text{furnace} - T_\text{reflector}), \text{ where htc} = \sigma F_\text{e} (T^2_\text{furnace} + T^2_\text{reflector})(T_\text{furnace} + T_\text{reflector}) && (17)
\end{aligned}$$

For this simulation, emissivities of magnesium oxide and steel are taken to be 0.18 and 0.35 respectively.

Finally, the reactor core has a 1 kW heat source distributed following a cosine shape axially and radially using a Moose function.

### 5.2.2 Results

The results in figure 12 show that the reactor is isothermal for all practical purposes. The recirculation at the reactor periphery has a minuscule impact in core temperature distribution. The fluid loses energy as it moves through the external pipe and later returns to the core to be re-heated.

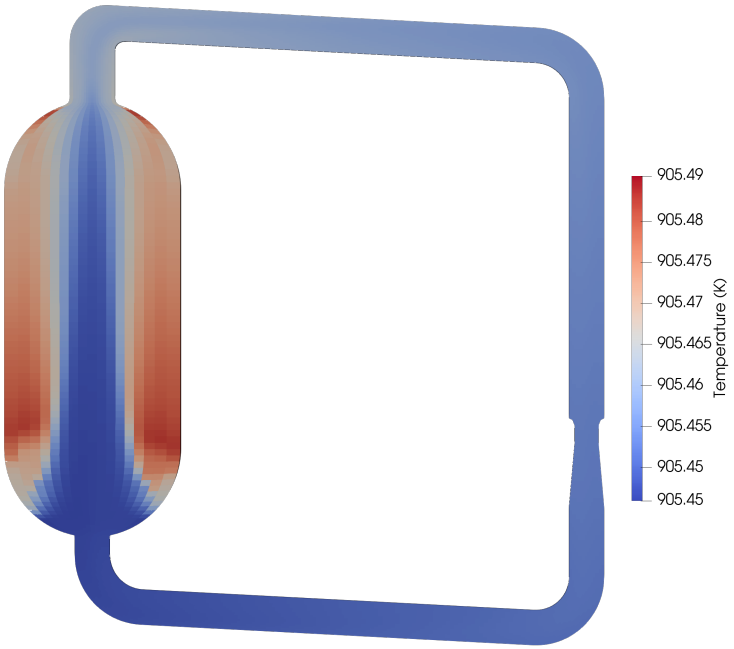



**Figure 12.** Fluid temperature field.

The reflector is heated by conjugate heat transfer from the core, diffuses energy isotropically and radiates it to the outside environment.

Heat transfer through the reflector is a relatively slow process, therefore a simulation restarting from the calibrated pressure drops and using the SIMPLE algorithm eliminating time derivatives of to ensure the asymptotic state was reached

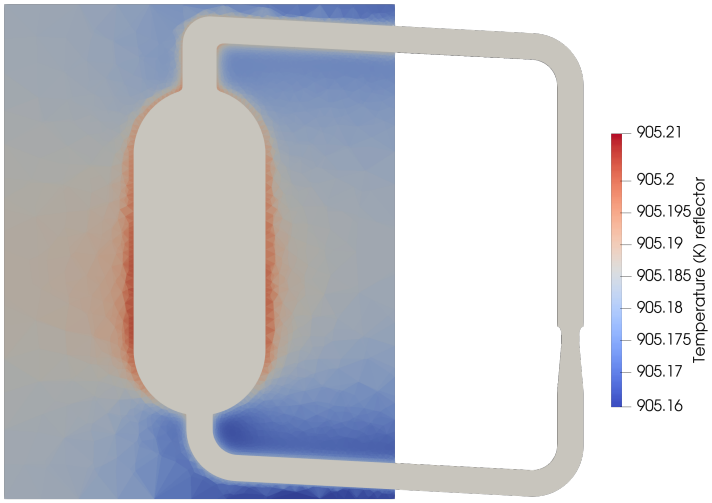



**Figure 13.** Reflector temperature field.

## 5.3 Precursors transport

Finally, delayed neutron precursor distributions are considered. This prepares the simulation to be coupled to neutronics by having an initial estimate of how precursors are distributed to accelerate convergence of the multiphysics case.

### 5.3.1 Geometry and Parameters

The geometry used is the same as the geometry of section 5.1.1. The parameters for production and destruction of precursors are seen in table 9, where Beta is the group-wise delayed neutron precursor production, and Lambda is the group-wise decay constant.

To model transport, turbulent momentum diffusivity is calculated first:

$$\nu_t = \frac{\mu_t}{\rho} \tag{18}$$

Calculation of the turbulent mass diffusivity $D_t$ follows equation 19, where it is obtained by rearranging the Schmidt number definition assuming a $Sc_t = 0.9$. It is then added to the laminar mass diffusivity ($D_l = 2.1 * 10^{-6}$ to obtain an effective one $D_{eff}$ as shown in equation 20. No good data is available for the turbulent mass transport of delayed neutron precursors.

**Table 9.** Delayed neutron precursor parameters.

| Group | Beta (-) | Lambda ($s^{-1}$) |
|---|---|---|
| 1 | 0.000250489 | 0.0124667 |
| 2 | 0.00104893 | 0.0282917 |
| 3 | 0.000726812 | 0.0425244 |
| 4 | 0.00143736 | 0.133042 |
| 5 | 0.0022503 | 0.292467 |
| 6 | 0.000680667 | 0.666488 |

$$D_{\mathsf{t}} = \frac{\nu_{\mathsf{t}}}{\mathrm{Sc}_{\mathsf{t}}} \tag{19}$$

$$D_{\mathsf{eff}} = D_{\mathsf{t}} + D_{\mathsf{l}} \tag{20}$$

### 5.3.2 Results

We can see the results of the precursor transport in the figures below. In figure 14 the longest lived precursor is shown, and it is evident that it is relatively well distributed throughout the fuel circuit.

In comparison with the longest lived group, we can see in figures **??** and **??** that shorter lived groups increasingly retain the shape of power distribution.

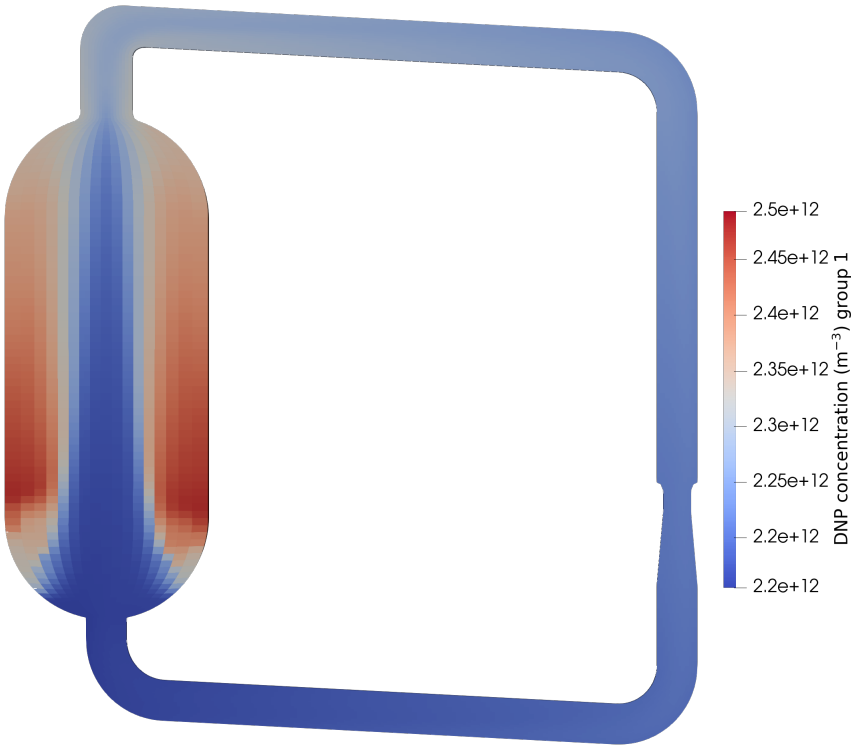



**Figure 14.** Delayed neutron precursor concentration in group 1.

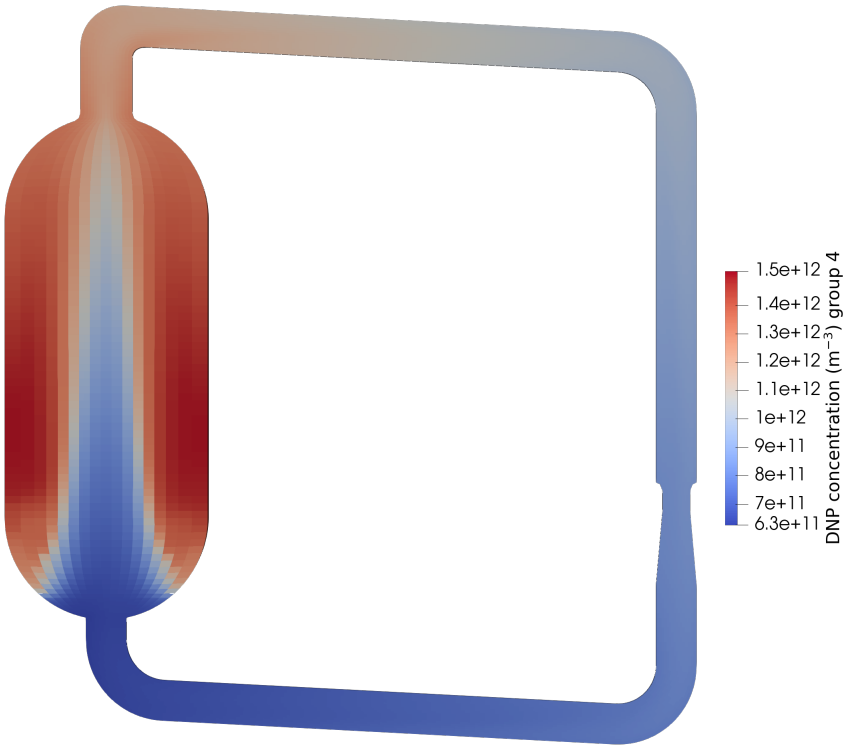



**Figure 15.** Delayed neutron precursor concentration in group 4.

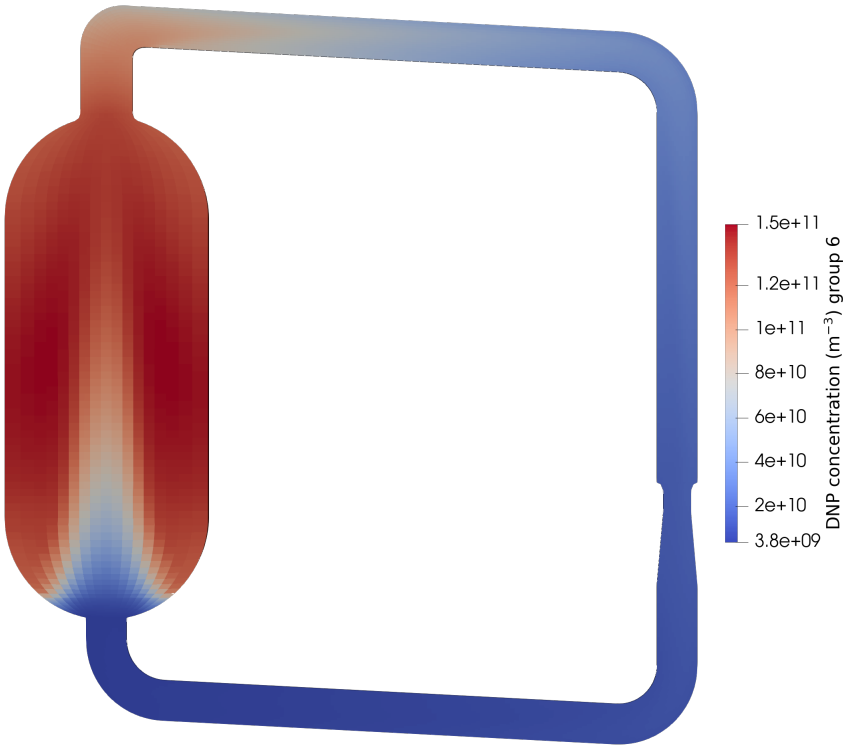



**Figure 16.** Delayed neutron precursor concentration in group 6.

# 6 CONCLUSIONS AND RECOMMENDATIONS

Closing remarks. Key findings of this ECAR::

- Finding 1.

Future actions

# 7 REFERENCES